

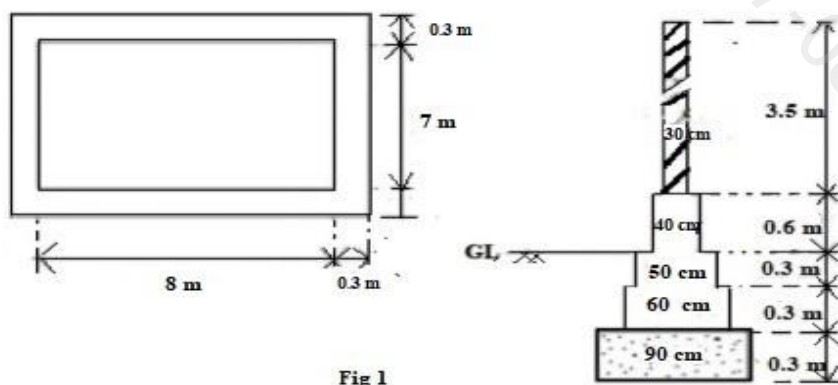
Supplementary Summer Examination – 2024**Semester: VII**

Subject Code: BTCVC704

Duration: 3 Hr.

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		12
A) Explain the significance of accurate estimation in construction projects. How does it impact project planning and execution?	CO1	6
B) Define specifications in the context of civil engineering projects and discuss their general purpose.	CO1	6
C) Fig 1. Shows the plan of superstructure wall (0.3 m) of a single room building of 8m x 7m, and sections represent the cross-sections of the walls with foundation. Estimate (Long wall and short wall method) the quantities of 1. Earthwork in excavation in foundation.(2 Marks). 2. Concrete in foundation.(2 Marks). 3. Brickwork in superstructure.(2 Marks).	CO1	6



Q.2	Solve Any Two of the following.	12
A)	Define rate analysis in the context of civil engineering and explain its significance in project cost estimation.	CO2 6

B)	Find out the materials required for the 20 Cu.m RCC work (1:1.5:3) excluding Steel.	CO2	6
C)	Discuss the factors that contribute to price escalation in civil engineering projects over time.	CO2	6
Q. 3	Solve Any Two of the following.		12
A)	Define a tender in the context of civil engineering projects and outline its primary purpose.	CO3	6
B)	Describe the key components typically included in a tender document for a civil engineering project.	CO3	6
C)	Explain the concept of project financing in B.O.T projects and how it differs from traditional procurement methods.	CO3	6
Q.4	Solve Any Two of the following.		12
A)	Explain the essentials of contracts briefly.	CO3	6
B)	Explain the methods employed by the Public Works Department (P.W.D) for planning and executing civil engineering projects.	CO3	6
C)	Explain the types of disputes commonly referred to arbitration in civil engineering projects and how they are resolved.	CO3	6
Q. 5	Solve Any Two of the following.		12
A)	What is valuation? Discuss its purpose. (2+4) Marks	CO4	6
B)	Describe the primary duties and responsibilities of a valuer when assessing civil engineering projects.	CO4	6
C)	A building having construction cost of Rs:15,000,00 has been constructed on a 150 m ² freehold land in metro city. The rate of land in the neighbourhood is Rs 3000/m ² . The outstanding such as sinking fund and taxes collectively is Rs:30000/annum.	CO4	6
	1. Determine the net rent of the property considering 6% cost of construction and 4% of the cost of land. (4 Marks)		
	2. Determine gross rent of the property after considering the outgoing (annually and monthly). (2 Marks)		

*** End ***